



Drag in Fluids

Water and air drag are complex phenomena. In swimming, swimmers aim to maximize the drag, and thus the propulsion. We aim to design a similar set of movement as that of a swimmer's hand, to reproduce and measure this drag.

The drag crisis has long been investigated (2); however, the research is usually on simple shapes such as cylinders and spheres. It partly shows that the drag can decrease as the Reynolds number, that is, the speed, increases; however, we believe this effect to be an unreachable ideal for more complex and human-like shapes.

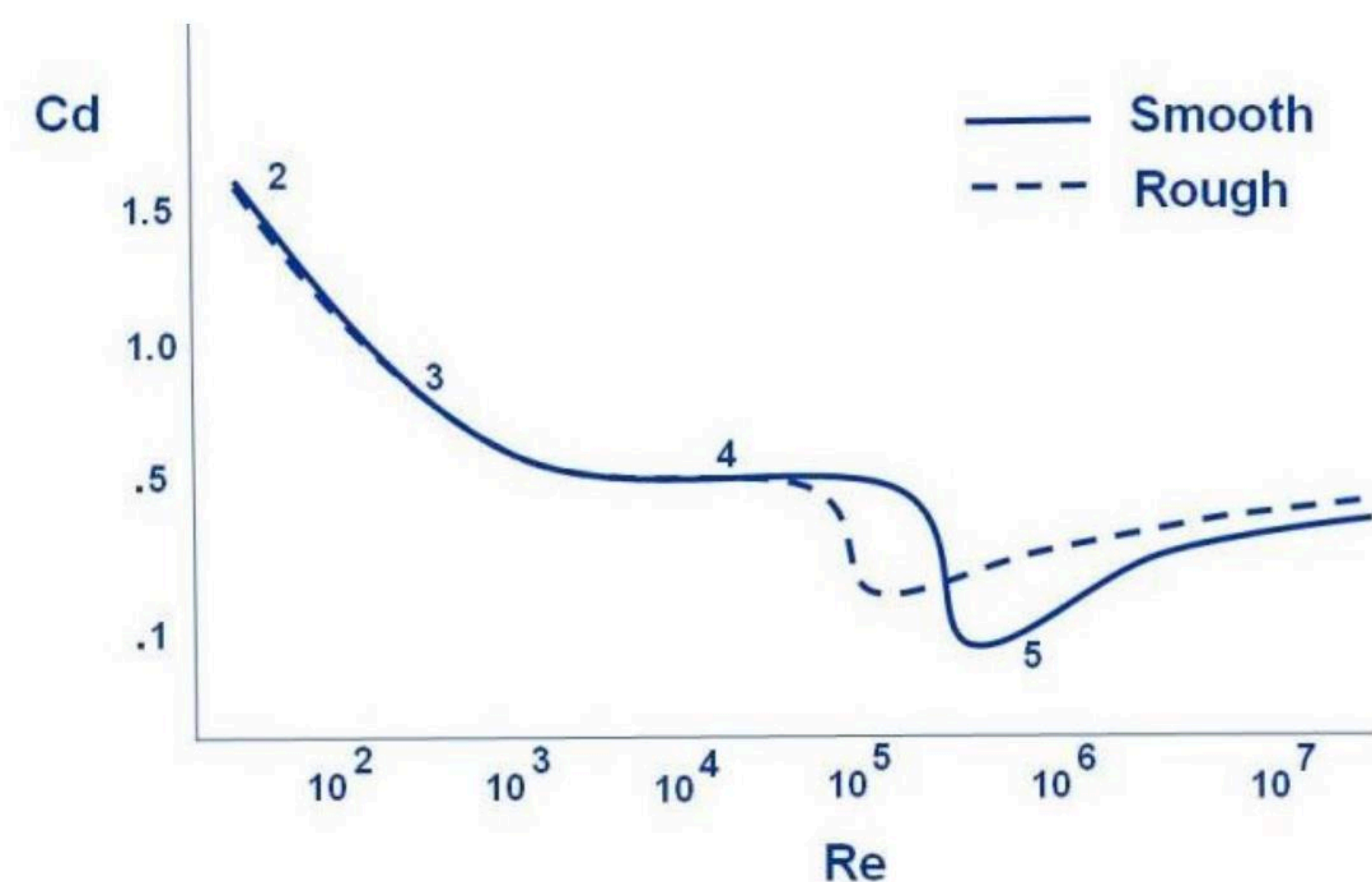


Figure I. Drag coefficient (C_d) versus Reynolds number (Re) for smooth and rough spheres. Surface roughness shifts the drag crisis to lower Reynolds numbers, reducing drag at typical speeds. Here, C_d is proportional to the drag force, and $Re = \rho UL / \mu$ with viscosity $0.01 \text{ cm}^2/\text{s}$.

Adapted from NASA

Methodology

We compute differences in drag by using pitot tubes around the equator of the hand, as well as select positions in between the fingers. We vary the distance between the fingers and measure the pressure force felt by each distribution, aiming to find a position that optimizes the drag. Pressure changes will be reflected as changes in water level in an arrangement of glass tubes that will be connected to each pitot tube, as shown in Figure II. A preliminary model was constructed using a simpler sphere model to test the effectiveness of this methodology.

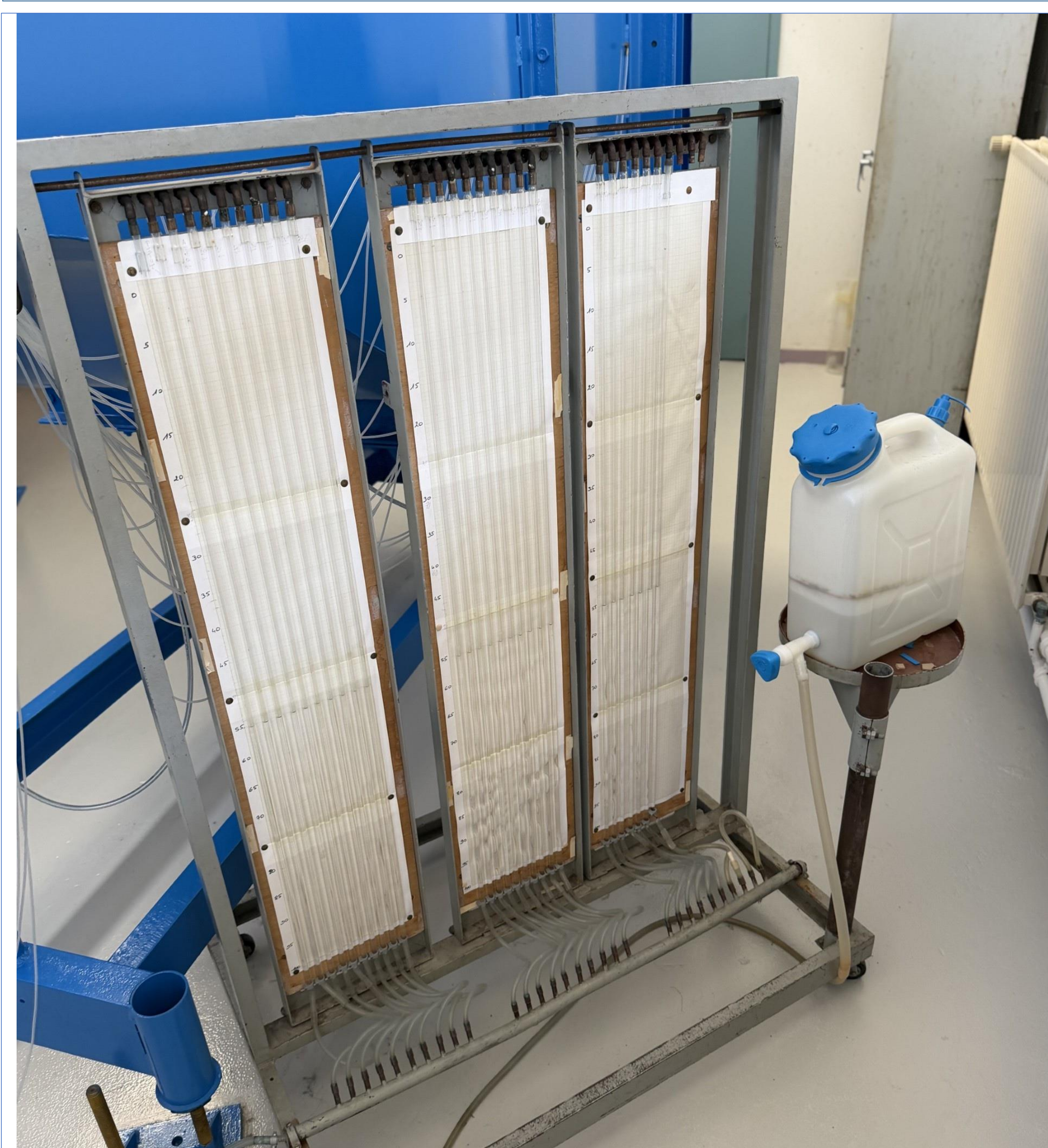


Figure II. Device that measures the pressure forces stimulated at all Pitot tubes.

The Hand Geometry

We aim to investigate the force exerted in each part of the hand, as well as the drag, so as to find a hand position that maximizes the force. Hand will be 3D printed in different configurations with holes bored at strategic places on its surface to connect pitot tubes to and measure the drag. We know empirically from swimmers that a flat hand should have the fingers slightly open, but we plan on exploring the specifics on how this happens and what is the optimal configuration of finger separation and angle between the wrist and knuckles.

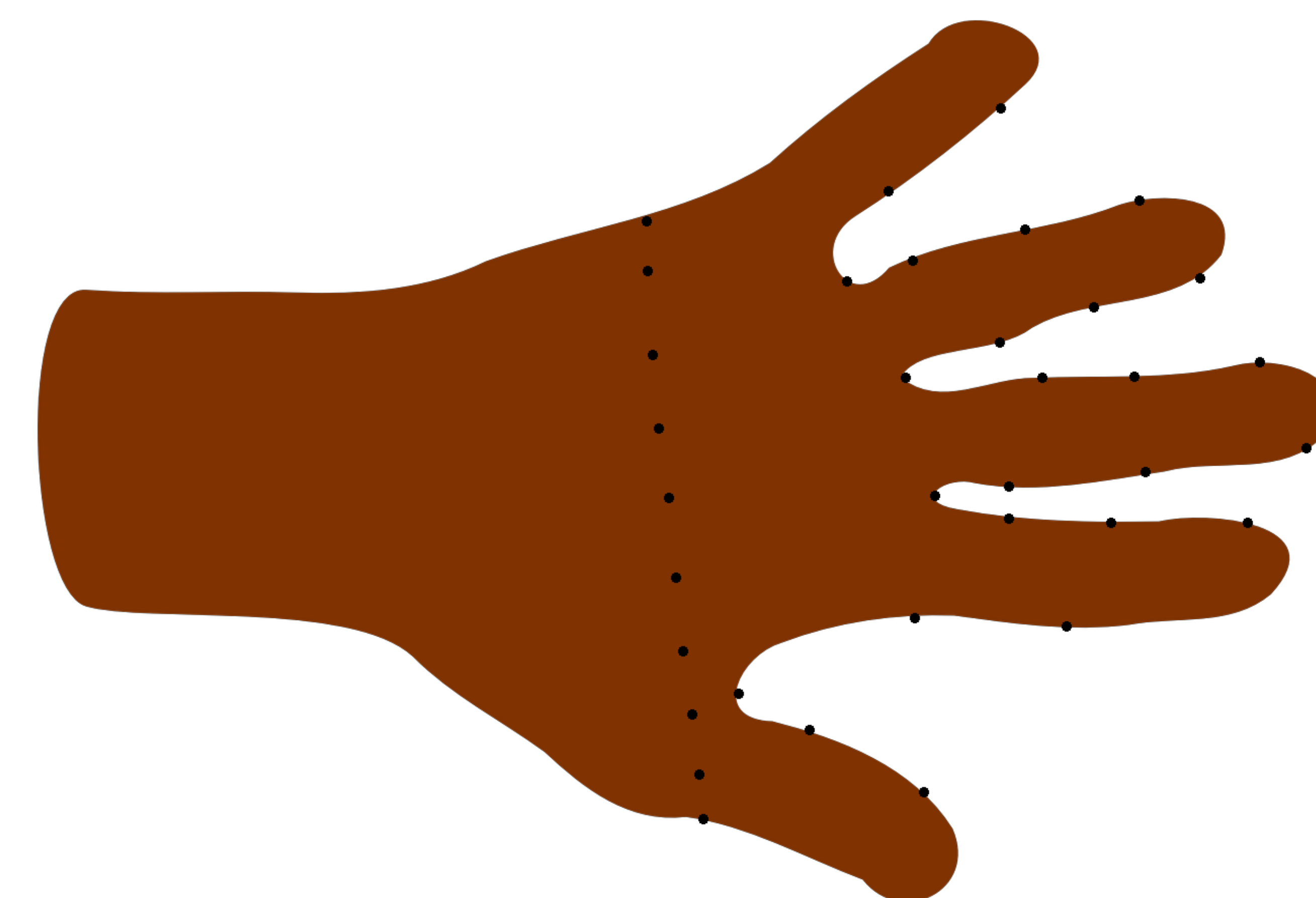


Figure III. Top view of a hand, with black dots in the locations where the pitot tubes are connected

References

1. NASA Glenn Research Center, *Drag of a Sphere*
2. Achenbach, E., *Experiments on the Flow Past Spheres at Very High Reynolds Numbers* (Journal of Fluid Mechanics, 1972)

Acknowledgements

1. Joint Fluids Lab
2. Professors Richard McLaughlin, Roberto Camassa and Jim Mahaney